

CURRENT ONE-STAGE AND TWO-STAGE IPDE DESIGN WITH EFFICACY

shared- α multicycle default model and TITE extension

Purpose. This document specifies the current efficacy-enabled AIDE/IPDE design. The shared- α multicycle additive model is the default for both toxicity and efficacy. The non-TITE design is the base trial; the TITE version preserves the same allocation, stopping, futility, recycling, MTD, and OBD rules while adding partial-follow-up likelihoods, an event clock, a waiting queue, and an n_{eval} decision-availability gate.

Design item	Current specification
Design family	AIDE/IPDE with binary toxicity and binary efficacy
Trial structures	One-stage and two-stage
Default toxicity model	shared- α multicycle additive CRM
Default efficacy model	shared- α multicycle additive β -binomial model
Toxicity stopping	No dose-by-dose toxicity elimination; terminate only when dose 1 is unsafe
Efficacy futility	Dose-specific Beta-Binomial posterior with Beta(1,1) prior
IPDE toxicity check	Patient-specific multicycle posterior risk at the next assigned dose
IPDE efficacy-benefit gate	Not used in the current design
Recycle destination	The next dose assigned by the trial; no search over alternative destinations
Recycle source dose	Any structurally eligible patient, including a patient already at the assigned dose
TITE principle	Same design rules as non-TITE plus TITE/delayed-outcome timing machinery

Document date: August 2026. This revision supersedes earlier descriptions that used a random-r individual IPDE gate, an efficacy-improvement recycle gate, flexible dose-direction controls, or dose-by-dose toxicity elimination.

Design Summary

Core principle. The toxicity model defines the current MTD and the dose-1 early-stop condition. Efficacy futility removes doses dose-by-dose. Utility ranks doses only after the MTD ceiling and persistent efficacy-futility exclusions are applied. A recycled patient is evaluated only for the dose already selected for the next cohort.

Current default model matrix

Component	Current rule	Key quantity
Toxicity	shared- α multicycle additive CRM	$S_{T,ic} = p_{T,dic} + \alpha_T S_{T,i,c-1}; q_{T,ic} = \min(1, S_{T,ic})$
Efficacy	shared- α multicycle additive model	$S_{E,ic} = p_{E,dic} + \alpha_E S_{E,i,c-1}; q_{E,ic} = \min(1, S_{E,ic})$
Individual recycle safety	Multicycle patient-specific posterior check	$\Pr(q_{T,next} > \varphi_T \mid D) < c_T$
Efficacy futility	Independent Beta(1,1) dose-level monitoring	$\Pr(p_{E,j} < \varphi_E \mid y_{j,n_j}) > c_E$
Recycle benefit	No posterior efficacy-benefit gate	Previous cycle must have no efficacy response
Destination	Use next assigned cohort dose	No alternative-destination search

Notation

Symbol	Definition	Symbol	Definition
i	Administration-level index	j	Dose-level index
d_{ic}	Dose actually received by patient i in cycle c	$p_{T,j}$	Regular toxicity probability at dose j
$p_{E,j}$	Regular efficacy probability at dose j	$S_{T,ic}$	Uncapped multicycle toxicity state
$S_{E,ic}$	Uncapped multicycle efficacy state	$q_{T,ic}$	Administration-specific toxicity probability
$q_{E,ic}$	Administration-specific efficacy probability	φ_T	Target toxicity probability
φ_E	Minimum efficacy threshold for futility	c_T, c_E	Posterior monitoring cutoffs
$w_{T,i}$	TITE toxicity follow-up weight	$w_{E,i}$	Efficacy follow-up weight
$\Delta_{E,i}$	Efficacy ascertainment indicator	R_j	1 if an observed efficacy response has occurred at dose j

1. Statistical Model

1.1 Baseline toxicity CRM

Let p_{0j} be the prespecified CRM skeleton probability at dose j and β the one-dimensional CRM parameter. The regular toxicity curve uses a power CRM:

$$p_{T,j}(\beta) = p_{0j}^{\exp}(\beta), \quad \beta \sim \text{Normal}(\mu_\beta, \sigma_\beta^2).$$

The regular toxicity probabilities $p_{T,j}$ are the quantities used for the MTD, utility, and the dose-1 safety stopping rule. Recycled administrations enter the likelihood through the multicycle probability defined below.

1.2 Baseline efficacy model

Regular efficacy is modeled independently by dose:

$$p_{E,j} \sim \text{Beta}(a_{E,j}, b_{E,j}), \quad j = 1, \dots, J.$$

The default implementation retains one row per administration because recycled observations depend on the complete same-patient cycle history.

1.3 shared- α multicycle carryover

For patient i with actual dose sequence $d_{i,1}, \dots, d_{i,C_i}$, define uncapped deterministic states. The same recursive form is used for toxicity and efficacy, with endpoint-specific α parameters.

$$\begin{aligned} S_{T,i_1} &= p_{T,d_{i_1}}; & S_{T,ic} &= p_{T,d_{ic}} + \alpha_T S_{T,i,c-1}, \quad c \geq 2. \\ q_{T,ic} &= \min(1, S_{T,ic}). \\ S_{E,i_1} &= p_{E,d_{i_1}}; & S_{E,ic} &= p_{E,d_{ic}} + \alpha_E S_{E,i,c-1}, \quad c \geq 2. \\ q_{E,ic} &= \min(1, S_{E,ic}). \end{aligned}$$

No intermediate cap. $S_{T,ic}$ and $S_{E,ic}$ are deterministic uncapped history states. The cap $\min(1, \cdot)$ is applied only to the probability used for the current Bernoulli outcome or proposed next-cycle risk. Replacing $S_{i,c-1}$ by $\min(1, S_{i,c-1})$ would define a different model.

1. Statistical Model (continued)

1.4 Geometric history expansion

The recursion implies geometric decay of prior-cycle contributions. For toxicity:

$$S_{T,ic} = p_{T,d,c} + \alpha_T p_{T,d,c-1} + \alpha_T^2 p_{T,d,c-2} + \dots + \alpha_T^{c-1} p_{T,d,1}.$$

The efficacy expression is identical after replacing T by E. In particular, for cycle 3:

$$\begin{aligned} q_{T,i_3} &= \min(1, p_{T,d_3} + \alpha_T p_{T,d_2} + \alpha_T^2 p_{T,d_1}), \\ q_{E,i_3} &= \min(1, p_{E,d_3} + \alpha_E p_{E,d_2} + \alpha_E^2 p_{E,d_1}). \end{aligned}$$

1.5 Priors for carryover parameters

The shared carryover coefficients are assigned prespecified Beta priors directly:

$$\alpha_T \sim \text{Beta}(a_T, b_T), \quad \alpha_E \sim \text{Beta}(a_E, b_E).$$

1.6 Complete-outcome likelihood

In the non-TITE trial, every administration contributes a Bernoulli likelihood term. For patient i and cycle c :

$$Y_{T,ic} / \Theta_T \sim \text{Bernoulli}(q_{T,ic}), \quad Y_{E,ic} / \Theta_E \sim \text{Bernoulli}(q_{E,ic}).$$

At cycle 1, $q_{T,i_1} = p_{T,d_1}$ and $q_{E,i_1} = p_{E,d_1}$. The full likelihood is a product over all administration rows, so a patient with two cycles contributes two likelihood terms, a patient with three cycles contributes three, and so on.

Two-cycle reduction. With $\text{cycle_max} = 2$, the multicycle model reduces exactly to the previous shared additive model: $q_{i_2} = \min(1, p_{\text{current}} + \alpha p_{\text{previous}})$, provided the same priors and all other design rules are used.

1. Statistical Model (continued)

1.7 True toxicity and efficacy generation

Simulation data generation follows the same multicycle recursion as the fitted model. The true state is carried forward patient-by-patient and remains uncapped.

$$\begin{aligned} S_{T,i_l}^{true} &= p_{T,d_{i_l}}^{true}; & S_{T,ic}^{true} &= p_{T,d_c}^{true} + \alpha_T^{true} S_{T,i,c-1}^{true}. \\ S_{E,i_l}^{true} &= p_{E,d_{i_l}}^{true}; & S_{E,ic}^{true} &= p_{E,d_c}^{true} + \alpha_E^{true} S_{E,i,c-1}^{true}. \\ Pr(DLT_{ic}=1) &= \min(1, S_{T,ic}^{true}); & Pr(EFF_{ic}=1) &= \min(1, S_{E,ic}^{true}). \end{aligned}$$

Therefore, for a patient treated at doses 1, 2, and 3 in successive cycles:

$$\begin{aligned} Pr(DLT_3=1) &= \min(1, p_{T,3}^{true} + \alpha_T^{true} p_{T,2}^{true} + (\alpha_T^{true})^2 p_{T,1}^{true}). \\ Pr(EFF_3=1) &= \min(1, p_{E,3}^{true} + \alpha_E^{true} p_{E,2}^{true} + (\alpha_E^{true})^2 p_{E,1}^{true}). \end{aligned}$$

1.8 Posterior summary convention

Posterior means are the default point estimates for regular $p_{T,j}$ and $p_{E,j}$ in MTD and utility calculations. Patient-specific recycle safety is evaluated draw-by-draw rather than by plugging posterior means into the multicycle formula.

Separation of quantities. Regular $p_{T,j}$ and $p_{E,j}$ drive MTD, utility, and final OBD selection. Patient-specific $q_{T,next}$ is reserved for the individual IPDE safety check. Efficacy $q_{E,next}$ is not currently used for a benefit gate.

2. Non-TITE Trial Design

2.1 Toxicity monitoring and MTD

The shared- α multicycle CRM is refit after each evaluated cohort. The CRM regular-dose curve $p_{T,j}$ defines the current toxicity-model MTD and toxicity recommendation.

Dose-1-only toxicity stopping. The current design does not eliminate individual dose levels for toxicity. The trial terminates when the posterior probability that dose 1 exceeds the toxicity target is above the prespecified cutoff:

$$Pr(p_{T,1} > \varphi_T / D) > c_T \Rightarrow \text{terminate the trial.}$$

If this condition is not met, no higher dose receives a permanent toxicity-elimination flag. Toxicity still affects the current MTD and cohort recommendation through the fitted CRM curve.

2.2 Dose-specific efficacy futility

Futility is intentionally separate from the multicycle efficacy model used for utility. At dose j , let n_j be the number of fully observed efficacy outcomes and y_j the number of observed responses. Use the fixed Beta(1,1) monitoring prior:

$$p_{E,j} / y_{j,n_j} \sim \text{Beta}(1 + y_j, 1 + n_j - y_j).$$

$$\text{Dose } j \text{ is futile if } Pr(p_{E,j} < \varphi_E / y_{j,n_j}) > c_E.$$

A futile dose remains futile for the rest of the trial and is removed from future cohort allocation, IPDE destination, and final OBD selection. This is the only dose-by-dose elimination mechanism in the current design.

2. Non-TITE Trial Design (continued)

2.3 One-stage design

1. Update the toxicity and efficacy models and calculate the current toxicity-model MTD j_{MTD} . Update persistent dose-specific efficacy futility.
2. For each dose j , let $R_j = 1$ if at least one observed efficacy response has occurred at dose j .
3. Among non-futile doses at or below j_{MTD} , if any $R_j = 1$, let j_R be the lowest response-producing dose. The utility candidate interval is all non-futile doses from j_R through j_{MTD} .
4. If the response-defined interval is nonempty, choose the dose with the highest prespecified utility as the allocation target, breaking ties toward the lower dose.
5. If no dose at or below the MTD has produced an observed efficacy response, use the current MTD as the allocation target.
6. Apply the universal no-skipping rule: when the target is above the current dose, assign the first untried dose encountered on the path toward the target. If every intervening dose has been tried, assign the target. If target $\leq$ current dose, assign the target directly.
7. Repeat after every evaluated cohort.

2.4 Two-stage design

8. Stage I is toxicity-driven AIDE. Efficacy outcomes are collected and futility is monitored, but cohort assignment follows the toxicity recommendation subject to standing futility exclusions.
9. Stage I ends only after the cumulative Stage I administration count reaches $s1_Max$ and the next toxicity recommendation is stay at the current dose.
10. Stage II uses exactly the one-stage rule above after every evaluated cohort. There is no separate top-two randomization or Stage-II-specific highest-utility mode.
11. After every Stage II cohort, refit the models, refresh the MTD, response indicators, futility state, utilities, and no-skipping-constrained allocation.

2. Non-TITE Trial Design (continued)

2.5 IPDE structural eligibility and assigned destination

A patient may be considered for another cycle when all structural conditions below are satisfied:

- The most recent administration has completed its toxicity and efficacy assessment.
- No DLT occurred in that completed administration.
- No efficacy response occurred in that completed administration.
- The patient has not reached cycle_max.
- The dose selected for the next cohort is not efficacy-futile and the trial has not stopped for dose-1 toxicity.

No source-dose restriction. Any structurally eligible patient may be recycled, regardless of the dose previously received. Same-dose recycling is allowed. A patient previously treated below, at, or above the newly assigned dose can be considered.

The patient is evaluated only for the dose already assigned to the next cohort. There is no search across alternative destinations and no flexible/upward-only destination parameter in the current design.

2.6 Individual multicycle toxicity-risk criterion

For each candidate patient and the next assigned dose k , reconstruct that patient's posterior uncapped state for every MCMC draw. If the patient's current posterior state is S_{current}^m , calculate:

$$q_{T,\text{next}}^m = \min(1, p_{T,k}^m + \alpha_T^m S_{\text{current}}^m).$$

$$\text{Allow recycling if } \text{mean}[\mathbb{1}\{q_{T,\text{next}}^m > \varphi_T\}] < c_T.$$

For a history $D1 \rightarrow D2$ and proposed assigned dose $D3$, this becomes $q_{T,3}^m = \min(1, p3^m + \alpha^m p2^m + \alpha^{m2} p1^m)$.

No efficacy-benefit gate. The current design does not require $E\{q_{E,\text{next}} - p_{E,\text{current}} \mid D\} > \delta$. Efficacy affects recycling through the structural requirement of no prior response and through dose-level futility only.

2. Non-TITE Trial Design (continued)

2.7 Enrollment and cohort filling

The non-TITE simulator may use either continuous enrollment with new-patient priority or an IPDE-first recruitment scheme. These mechanisms determine which eligible person fills an available cohort position; they do not change the cohort dose selected by the one-stage/two-stage algorithm.

Scheme	Queue priority	Operational rule
Continuous enrollment	Waiting new patients first	Enrollment remains open; waiting new patients fill cohort positions before eligible IPDE patients.
IPDE-first recruitment	Eligible IPDE patients first	Eligible IPDE patients fill first; recruit only the remaining number of new patients needed to complete the cohort.

2.8 Final OBD selection

12. Obtain the final toxicity-model MTD.
13. Exclude doses that are persistently efficacy-futile.
14. Require the dose to have been tried at least once.
15. Require at least one observed efficacy response at the dose.
16. Require the dose to be at or below the final MTD.
17. Among eligible doses, select the highest-utility dose; break ties toward the lower dose. If no dose satisfies all conditions, report no OBD.

$$C_{OBD} = \{ j : \text{tried}_j = 1, R_j = 1, \text{futile}_j = 0, j \leq j_{MTD} \}.$$

3. TITE Extension of the Same Trial

Design-preservation rule. The TITE trial is the non-TITE design plus time-to-event handling. TITE does not introduce a different one-stage rule, two-stage rule, futility rule, final OBD rule, recycle destination rule, or toxicity-elimination rule.

3.1 Toxicity TITE likelihood with multicycle probability

At interim time t , let $w_{T,i}$ be the toxicity follow-up fraction for administration i . For an observed DLT, $w_{T,i}$ is forced to one. The pre-TITE probability is the administration-specific multicycle probability $q_{T,i}$. The TITE event probability is:

$$\theta_{T,i}^{eff} = w_{T,i} q_{T,i}$$

$$L_{TITE,T} = \prod_i (w_{T,i} q_{T,i})^{Y_{ti}} [1 - w_{T,i} q_{T,i}]^{1-Y_{ti}}.$$

Toxicity status	Weight	Likelihood contribution
Observed DLT	$w_T = 1$	q_T
Completed non-DLT	$w_T = 1$	$1 - q_T$
Pending non-DLT	$0 < w_T < 1$	$1 - w_T q_T$
No follow-up	$w_T = 0$	1

3.2 Efficacy delayed-outcome likelihood

Let $\Delta_{E,i} = 1$ when efficacy is ascertained by interim time t and 0 for a pending nonresponse. The row-specific efficacy probability is $q_{E,i}$ from the multicycle model. The delayed-outcome approximation uses:

$$\log L_{E,i} = \Delta_{E,i} Y_{E,i} \log(q_{E,i}) + [\Delta_{E,i}(1 - Y_{E,i}) + (1 - \Delta_{E,i})w_{E,i}] \log(1 - q_{E,i}).$$

3. TITE Extension (continued)

3.3 TITE efficacy futility remains Beta-Binomial(1,1)

The delayed-outcome efficacy model is used for utility estimation, but dose-level futility remains the separate Beta-Binomial monitoring rule. Only ascertained efficacy outcomes enter n_j and y_j ; pending outcomes do not contribute to the futility counts.

$$p_{E,j} / \text{ascertained data} \sim \text{Beta}(1+y_j, 1+n_{j-y_j}).$$

$$\text{futile}_j = 1 \text{ if } \Pr(p_{E,j} < \varphi_E / \text{ascertained data}) > c_E.$$

3.4 TITE toxicity stopping remains dose-1-only

TITE partial follow-up changes the posterior distribution of $p_{T,j}$, but it does not create dose-by-dose toxicity elimination. At every TITE decision:

$$\Pr(p_{T,i} > \varphi_T / \text{current TITE data}) > c_T \Rightarrow \text{terminate}.$$

Otherwise, the fitted TITE-CRM curve is used for the current MTD and toxicity recommendation. Higher doses are not persistently eliminated for toxicity.

3.5 Patient-specific TITE recycle safety

For a waiting retreat patient, use the posterior draws from the current multicycle TITE toxicity fit. Reconstruct the uncapped state for the source administration identified by `source_admin_id`, then evaluate the next assigned dose k :

$$q_{T,next}^m = \min(1, p_{T,k}^m + \alpha_T^m S_{source}^m).$$

$$\text{Allow if } \Pr(q_{T,next} > \varphi_T / \text{current TITE data}) < c_T.$$

This is the same individual-risk rule as the non-TITE trial; the difference is that the posterior is based on partial toxicity follow-up.

3. TITE Extension (continued)

3.6 Event clock and waiting queue

The TITE trial tracks calendar time explicitly. Assignable events are:

- a new-patient arrival; or
- completion of an administration that creates a structurally eligible retreat opportunity.

Every assignable opportunity enters the waiting queue. The current TITE implementation uses new-patient-first priority.

3.7 Open cohort versus closed cohort

State	Operational rule
Open cohort	Do not make a new model-based dose decision. Fill available cohort positions from the queue using new-patient-first priority. Eligible retreat patients are checked for the cohort's already assigned dose.
No open cohort	Every assignable event first triggers the n_{eval} decision-availability check. If the gate is open, rebuild interim data, fit the TITE toxicity and efficacy models, apply the same one-stage/two-stage design, open the new cohort, and fill it from the queue.

3.8 n_{eval} decision-availability gate

Let $N_{eval}(d_{current})$ be the number of administrations at the current dose with fully ascertained DLT status, either because a DLT has occurred or the DLT window has completed without DLT.

If $N_{eval}(d_{current}) < n_{eval}$: no model-based decision is made and all waiting opportunities remain queued.

Once $N_{eval}(d_{current}) \geq n_{eval}$, the active one-stage or two-stage rule may be evaluated. The starting cohort is opened according to the initialization rule and is not blocked by this gate.

4. Utility Functions

Utilities use regular-patient probabilities $p_{E,j}$ and $p_{T,j}$. Carryover parameters affect these regular-probability posteriors through recycled-administration likelihood contributions, but patient-specific q values are not inserted directly into the utility.

4.1 Utility 1: efficacy only

$$U_1(j) = p_{E,j}.$$

4.2 Utility 2: efficacy minus toxicity penalty

$$U_2(j) = p_{E,j} - \lambda_T p_{T,j}, \quad \lambda_T \geq 0.$$

4.3 Utility 3: four-outcome score

$$U_3(j) = u_{00}(1-p_{T,j})(1-p_{E,j}) + u_{01}(1-p_{T,j})p_{E,j} + u_{10}p_{T,j}(1-p_{E,j}) + u_{11}p_{T,j}p_{E,j}.$$

Only doses in the active candidate set are ranked. The conservative default tie rule selects the lower dose among equal-utility candidates.

5. Control Parameters

Parameter	Role	Current convention
tox_model	Toxicity model	shared- α multicycle additive CRM
eff_model	Efficacy model	shared- α multicycle additive
α_T prior	Shared toxicity carryover	Prespecified Beta prior
α_E prior	Shared efficacy carryover	Prespecified Beta prior
φ_T, c_T	Toxicity target and cutoff	Dose-1 early stop and individual recycle safety
φ_E, c_E	Futility threshold and cutoff	Dose-specific Beta(1,1) efficacy futility
cycle_max	Maximum cycles per patient	User specified
s1_Max	Two-stage Stage-I transition threshold	User specified; transition also requires stay
n_eval	TITE decision availability	Fully evaluated DLT administrations at current dose
utility_type	Dose ranking	1, 2, or 3
enrollment	Queue/cohort filling	Non-TITE: selected scheme; TITE: new-patient-first event clock

6. Compact Algorithms

6.1 One-stage algorithm

18. Fit the shared- α multicycle toxicity and efficacy models.
19. If dose 1 is unsafe according to the posterior cutoff, terminate the trial.
20. Update persistent Beta(1,1) dose-specific efficacy futility.
21. Calculate the toxicity-model MTD.
22. If any non-futile dose at or below the MTD has an observed efficacy response, form the response-floor interval and select its highest-utility dose as the target; otherwise use the MTD.
23. Apply the universal upward no-skipping rule and assign the next cohort dose.
24. For each structurally eligible retreat patient considered for a cohort position, evaluate the patient-specific multicycle toxicity risk only at that assigned dose. If the risk passes, the patient can fill the position.
25. Repeat after every evaluated cohort.

6.2 Two-stage algorithm

26. Run toxicity-driven Stage I until the Stage-I administration threshold is reached and the next toxicity recommendation is stay.
27. Beginning in Stage II, use the one-stage algorithm above after every evaluated cohort.
28. At final analysis, select the OBD using the common tried + response + non-futile + at-or-below-MTD rule.

6.3 TITE algorithm

29. Use the same one-stage/two-stage design rules as the non-TITE trial.
30. Process new arrivals and retreat opportunities on the event clock and retain them in the waiting queue.
31. When a cohort is open, fill it without making a new dose decision.
32. When no cohort is open, require $N_{\text{eval}}(d_{\text{current}}) \geq n_{\text{eval}}$ before fitting models or making a decision.
33. Fit multicycle TITE toxicity and delayed-outcome efficacy models with partial follow-up.
34. Apply the same dose-1-only toxicity stopping, Beta(1,1) futility, MTD/utility allocation, no-skipping, assigned-dose retreat safety, and final OBD rules.

A. Implementation Templates

A.1 Complete-outcome multicycle toxicity

```
alpha_T <- g1 / (g1 + g2)
for (r in 1:N_admin) {
  S_T[r + 1] <- p_T[dose[r]] + alpha_T * S_T[previous_state_index[r]]
  q_T[r] <- min(1, S_T[r + 1])
  y_T[r] ~ dbern(q_T[r])
}
```

A.2 TITE multicycle toxicity

```
for (r in 1:N_admin) {
  S_T[r + 1] <- p_T[dose[r]] + alpha_T * S_T[previous_state_index[r]]
  q_T[r] <- min(1, S_T[r + 1])
  y_T[r] ~ dbern(w_T[r] * q_T[r])
}
# force w_T[r] = 1 for an observed DLT
```

A.3 Complete-outcome multicycle efficacy

```
for (r in 1:N_admin) {
  S_E[r + 1] <- p_E[dose[r]] + alpha_E * S_E[previous_state_index[r]]
  q_E[r] <- min(1, S_E[r + 1])
  y_E[r] ~ dbern(q_E[r])
}
```

A. Implementation Templates (continued)

A.4 Delayed-outcome multicycle efficacy

```
for (r in 1:N_admin) {
  S_E[r + 1] <- p_E[dose[r]] + alpha_E * S_E[previous_state_index[r]]
  q_E[r] <- min(1 - eps, max(eps, S_E[r + 1]))
  logLik[r] <- ascertain[r] * y_E[r] * log(q_E[r]) +
    (ascertain[r] * (1 - y_E[r]) + (1 - ascertain[r]) * w_E[r]) * log(1 - q_E[r])
  # implement -logLik[r] through a zeros trick or equivalent custom likelihood
}
```

A.5 Individual multicycle recycle toxicity screen

```
# source state is reconstructed draw-by-draw from the candidate patient history
q_next_draw <- pmin(1, p_draw[, next_dose] + alpha_draw * current_state_draw)
probability_over_phi <- mean(q_next_draw > phi_T)
allowed <- probability_over_phi < c_T
```

A.6 Beta-Binomial futility

```
posterior_j <- Beta(1 + y_j, 1 + n_j - y_j)
prob_below_j <- pbeta(phi_E, 1 + y_j, 1 + n_j - y_j)
futile_j <- (prob_below_j > c_E)
# in TITE, n_j and y_j include ascertained efficacy outcomes only
```

B. References

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- O'Quigley, J., Pepe, M., and Fisher, L. (1990). Continual reassessment method: a practical design for phase I clinical trials in cancer. *Biometrics* 46, 33-48.
- Lin, R., and Yuan, Y. (2020). Time-to-event model-assisted designs for dose-finding trials with delayed toxicity. *Biostatistics* 21(4), 807-824.